COSC264 · INTRODUCTION TO COMPUTER NETWORKS AND THE INTERNET · TERM 4 · MODULE 6

# Application Protocols: the Web, Email and DNS

1 · Network applications | 2 · The Web and HTTP | 3 · Email | 4 · DNS

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OUTLINE

# Module 6 — roadmap

- General concepts about network applications
- One network application: the Web
- HTTP: the protocol that supports the Web
- Email
- DNS: the Domain Name System

MODULE 6

# Network Applications

What an application needs from the layers beneath it

1 · Network applications | 2 · The Web and HTTP | 3 · Email | 4 · DNS

NETWORK APPLICATIONS

# What is a network application?

**Programs that run on different end systems** and communicate with each other over the network.

**Web browsing**

Google Chrome, Mozilla Firefox, Microsoft Edge

**Email**

Gmail, Outlook

**Instant messaging**

Slack, WhatsApp, Facebook Messenger

**Streaming services**

YouTube, Netflix

**Social media**

Instagram, TikTok

**AI assistants**

Claude, ChatGPT, Gemini

# Application-layer protocols

An **application-layer protocol** is a big piece of a network application.

It defines how the application's processes **pass messages**: which messages exist, their format, and when to respond — so e.g. any browser can talk to any web server.

| Network application | Application-layer protocol |
|---|---|
| Web application | HTTP |
| Email | SMTP |

To carry these messages, it relies on a **transport-layer protocol** (TCP or UDP).

# Choosing a transport protocol

- Application-layer protocols are controlled by the **developer**; transport-layer protocols are built into the **OS**.
- So the developer picks the OS transport protocol that best fits the application's needs:

| Property | Example applications | Transport protocol |
|---|---|---|
| **Reliable data transfer** | Email, IM, file transfer, financial apps | **TCP** preferred<br>UDP unsuitable |
| **Low delay** | Internet telephony, teleconferencing, multiplayer games | **UDP** preferred<br>TCP retransmissions add latency |

MODULE 6

# The Web and HTTP

The application-layer protocol behind every page you load

1 · Network applications | **2 · The Web and HTTP** | 3 · Email | 4 · DNS

THE WEB

# The World Wide Web

**THE WEB**

The **World Wide Web** (aka Web) is a **network application**: a global collection of documents and other resources shared over the Internet — like a large file storage system.

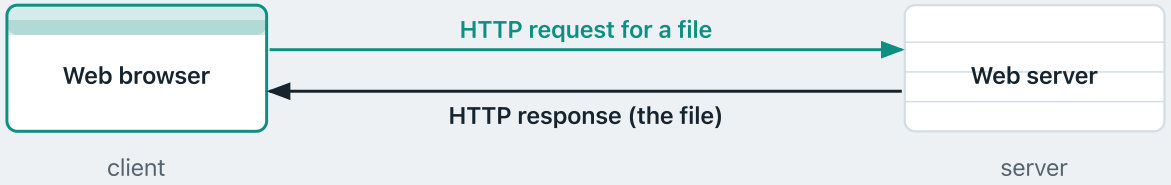
Each object on the Web (e.g., an image, an HTML file) is addressable by a **URL**:

```
http://www.someschool.edu/someDept/pic.gif
     └── host name ──┘ └──── path name ─────┘
```

# HTTP: the protocol behind the Web

**HTTP** | The **Hypertext Transfer Protocol** is the **application-layer protocol** that supports the Web — it defines how clients request objects by URL and how servers respond.

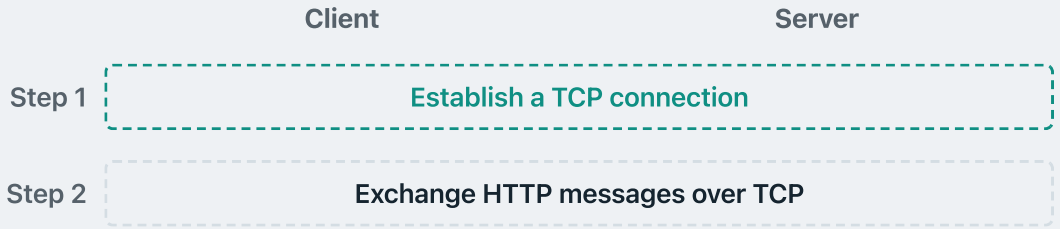
HTTP follows a **request–response** pattern: the client sends an **HTTP request** for an object, and the server answers with an **HTTP response** carrying it.



HTTP

# How HTTP works

HTTP uses **TCP** as its transport protocol — so before any HTTP message is sent, a TCP connection must be established first.




**STATELESS** HTTP is **stateless** — the server maintains no information about past client requests. Each request is handled independently.

# HTTP connections

### Non-persistent HTTP

- At most **one object** is sent over a TCP connection — connection is then closed
- HTTP/1.0 uses non-persistent HTTP

### Persistent HTTP

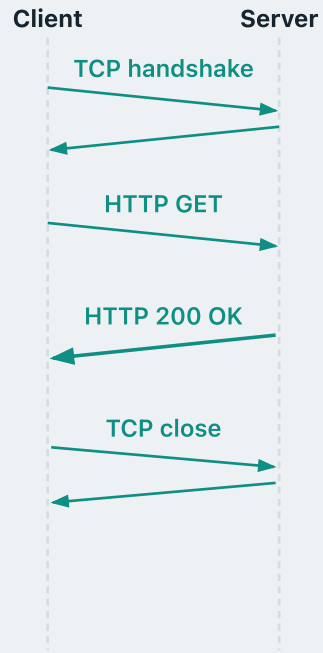
- **Multiple objects** can be sent over a single TCP connection
- HTTP/1.1, HTTP/2, and HTTP/3 use persistent HTTP

| Protocol | Published | Connection type | Status |
|---|---|---|---|
| **HTTP/1.0** | 1996 | Non-persistent | Legacy |
| **HTTP/1.1** | 1997 | Persistent | Standard |
| **HTTP/2** | 2015 | Persistent | Standard |
| **HTTP/3** | 2022 | Persistent (QUIC) | Standard |

# Non-persistent HTTP in action

Suppose a client wants to retrieve a file from: `http://www.someSchool.edu/someDepartment/home.index`

1. Client establishes a **TCP connection** with the server
2. Client sends an **HTTP GET** request — path: `/someDepartment/home.index`
3. Server retrieves the HTML file, wraps it in an **HTTP response**, and sends it back
4. Server signals TCP to **close the connection**



# Response time: non-persistent HTTP

**RESPONSE TIME** — The time to retrieve a single object using HTTP — from the client's request until the object is received

**Q:** What is the response time for non-persistent HTTP?

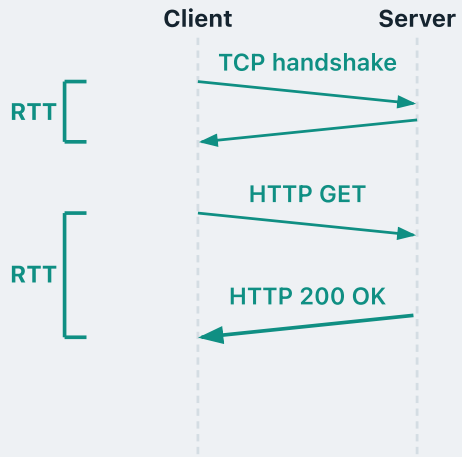
**A: 2 RTTs + file transmission time**

**RTT** — Recall: **round-trip time** — the time for a small packet to travel from client to server and back

**First RTT**: the TCP handshake — client sends SYN, server replies SYN-ACK

**Second RTT**: client sends HTTP GET, server sends back the response

With many referenced objects, this cost multiplies — a significant burden on the server



# Persistent HTTP in action

Client and server establish a TCP connection and exchange HTTP messages

Server **keeps the TCP connection open** after sending the first HTTP response

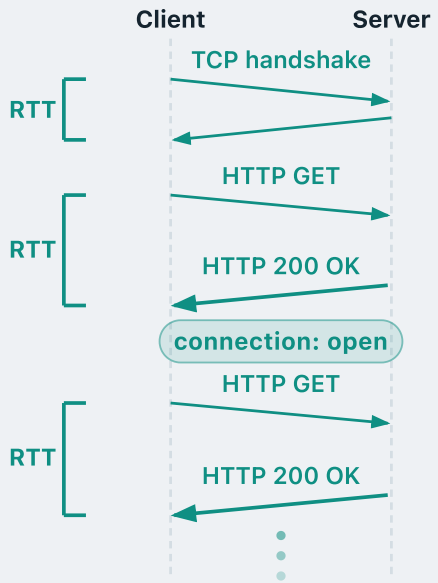
Subsequent HTTP messages between the same client and server use the **same TCP connection** — no new handshake needed

**Response time on one persistent connection:**

1 RTT for the initial TCP handshake, then 1 RTT for each HTTP request.

So for $n$ requests: **$\text{total} \approx (1 + n)$ RTT** (plus transmission time).

Amortised over the $n$ objects: $(1 + n)/n = (1/n + 1)$ RTT. As $n$ becomes large, $1/n \to 0$, so this approaches **~1 RTT per object**.

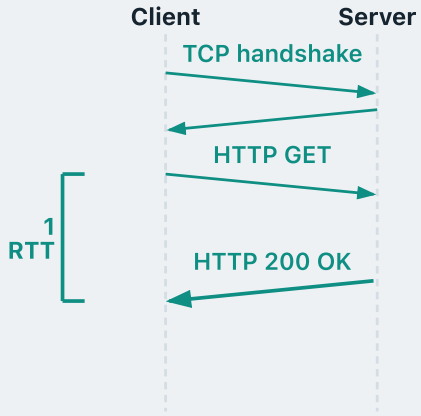



**Can we do even better than 1 RTT per object?**

# Improving HTTP efficiency further with pipelining

**WITHOUT PIPELINING**

Client issues a new request **only after** receiving the previous response — **1 RTT per file**



# Improving HTTP efficiency further with pipelining

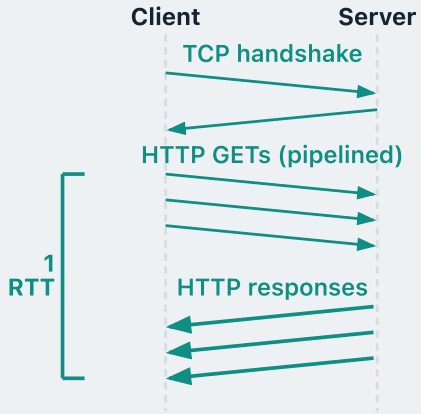
**WITHOUT PIPELINING**

Client issues a new request **only after** receiving the previous response — **1 RTT per file**

**WITH PIPELINING**

Client sends requests **as soon as it needs to retrieve an object**, without waiting for responses — e.g. in this example we pipeline 3 HTTP requests all together

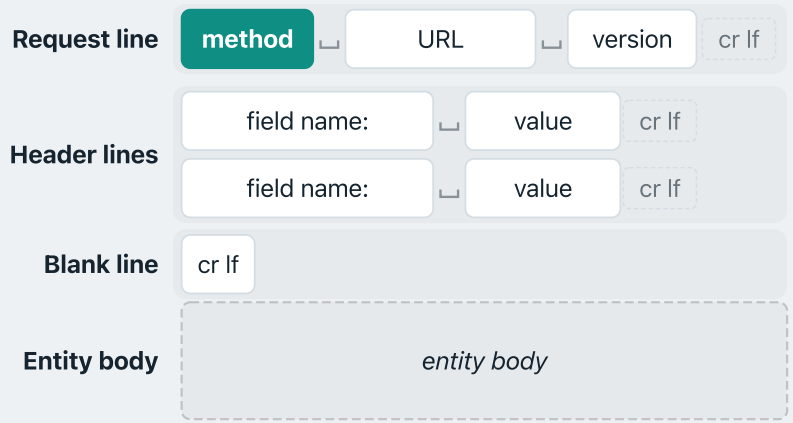
**Benefit**: pipelining 3 requests → response time per file ≈ **1 RTT / 3**



# HTTP request message format

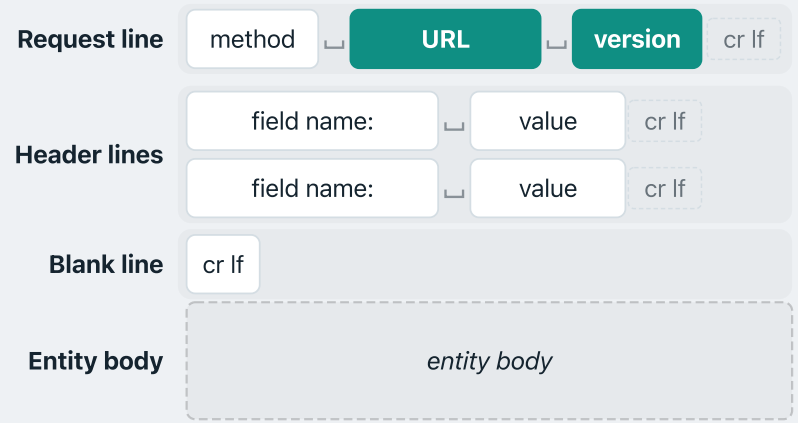
**Method — what action to perform**

| Method | Action |
|---|---|
| **GET** | **Retrieve a resource** |
| HEAD | Retrieve headers only (no body) |
| **POST** | **Submit data to a resource** |
| PUT | Store / replace a resource |
| **DELETE** | **Remove a resource** |



# HTTP request message format

**URL**: path to the requested resource — e.g. `/someDept/pic.gif`
**HTTP Version**: identifies the protocol in use — `HTTP/1.0` or `HTTP/1.1`



# HTTP request message format

### Header lines

- Zero or more `field-name: value` pairs, one per line
- Common examples:

```
Host: www.example.com
```

Mandatory in HTTP/1.1 — identifies the target server

```
Content-Type: text/html; charset=UTF-8
```

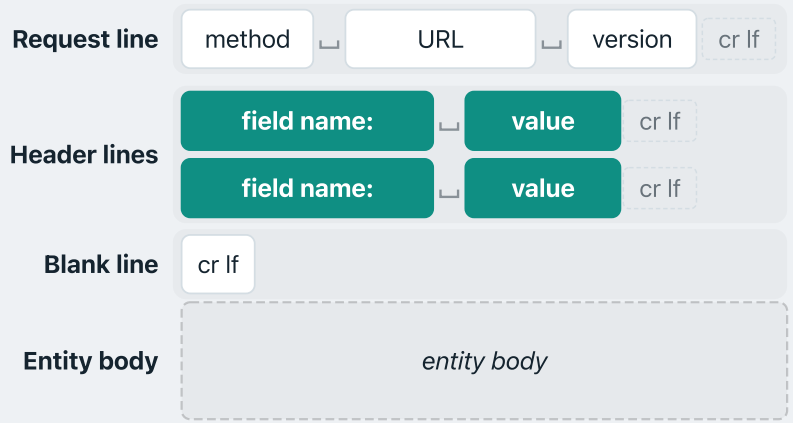
Format of the message body

```
Connection: close
```

Close the TCP connection after this response

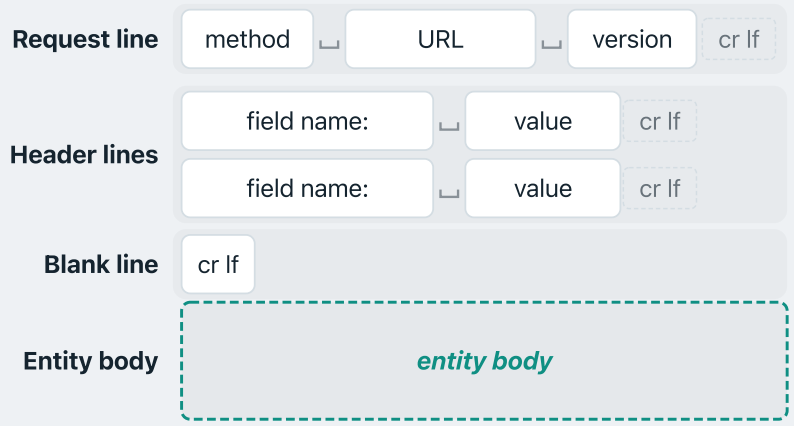
```
Accept-Language: da, en-gb;q=0.8,
en;q=0.7
```

Preferred languages, weighted by q (default 1): Danish → British English → any English



# HTTP request message format

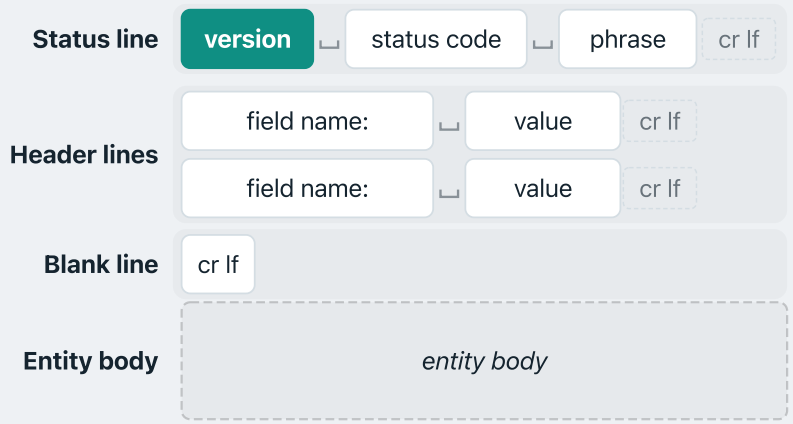
> **Entity body**: empty for `GET` and `HEAD`; carries the payload for `POST` and `PUT` (e.g. form data or JSON)



# HTTP response message format

**Version**

The HTTP version the server is using — e.g. `HTTP/1.1`, just as in the request line.

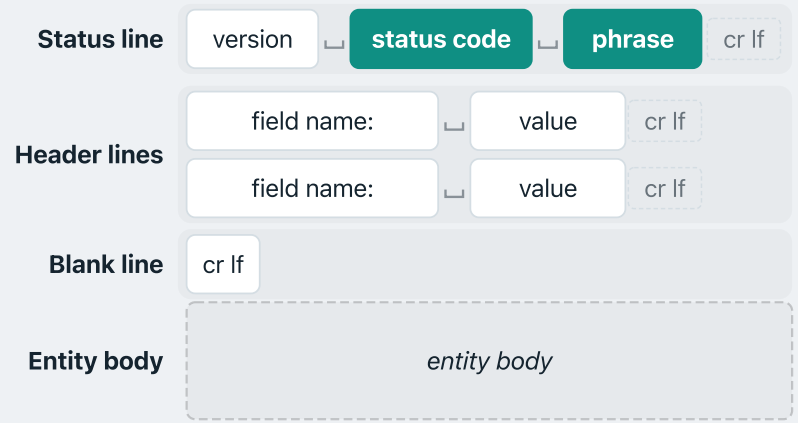


HTTP

# HTTP response message format

**Status code & phrase**

- **Status code**: 3-digit number indicating the result of the request
- **Phrase**: human-readable text accompanying the code — e.g. `OK`, `Not Found`

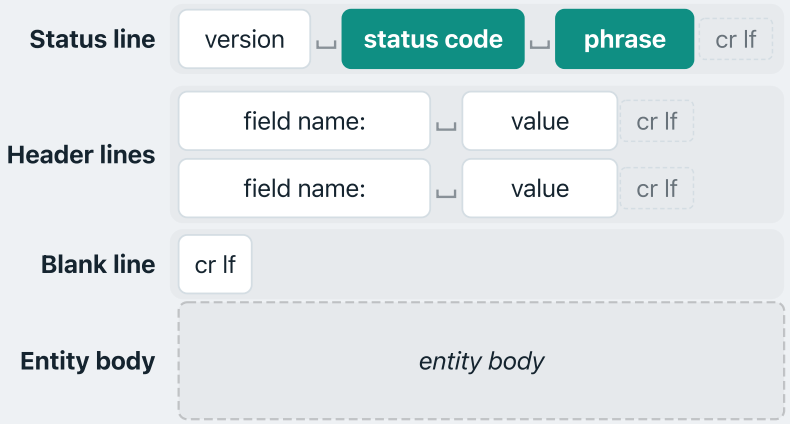


HTTP

# HTTP response message format

**Common examples**

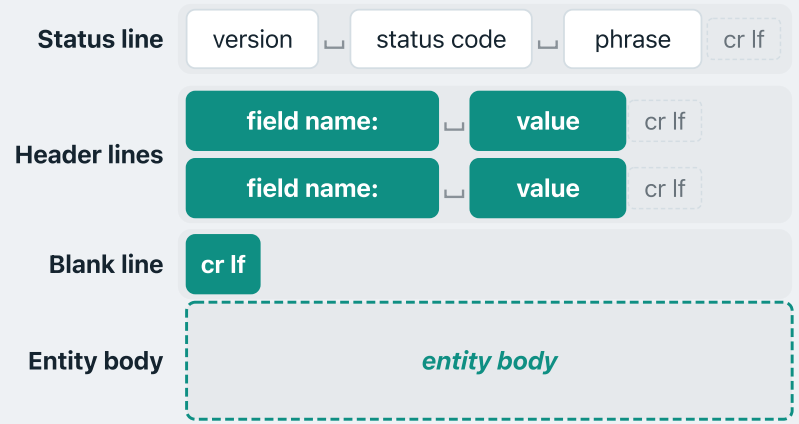
| Code | Phrase |
|---|---|
| 200 | OK |
| 301 | Moved Permanently |
| 400 | Bad Request |
| 404 | Not Found |
| 505 | HTTP Version Not Supported |



# HTTP response message format

**Header lines, blank line, entity body**

Same structure as in the request: `field-name: value` header lines, a blank line (a lone CRLF), then the body — here, the **requested object** itself.



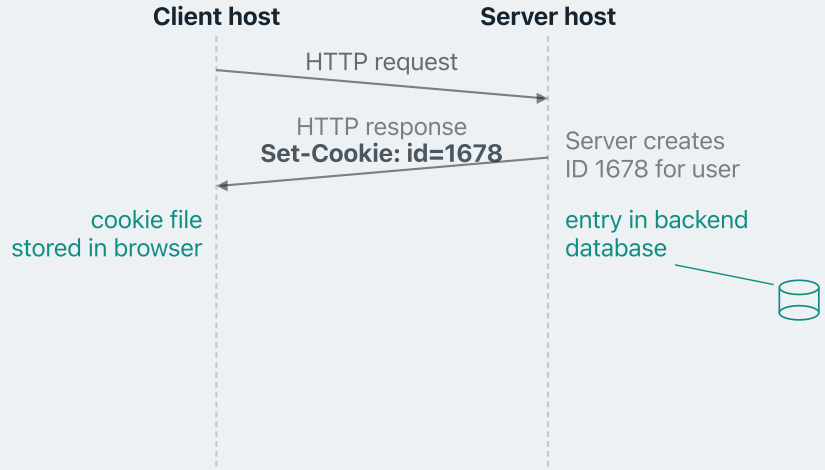
HTTP

# Cookies

> HTTP is **stateless** — every request is independent; the server retains no memory of past requests.
>
> So the server does not remember your earlier conversations with it.
>
> Yet many applications *need* that memory: e.g. refreshing an online shop's page sends a **new HTTP request**, and the items in your cart should not just disappear.
>
> **How does the server know what is in your cart?**

The answer: **cookies**.

# Cookies: an example [RFC 6265]




The client sends an HTTP request to the host

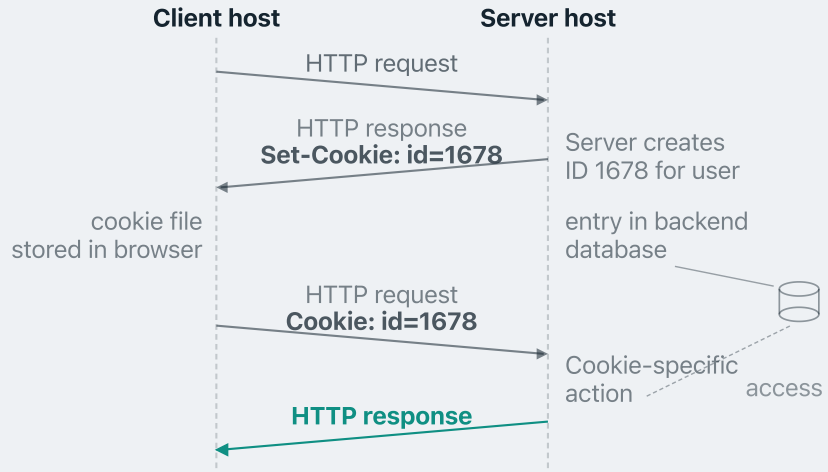
The server generates an **HTTP response**; on top of that, it prepares a **cookie** for the user.

The cookie uniquely identifies this session with the client, e.g. **id=1678**.

It is embedded in the response as a header line: **Set-Cookie: id=1678**.

The ID is stored as an entry in the server's **back-end database**, and in the client's **cookie file**, managed by the browser

# Cookies: an example [RFC 6265]




Later, the client sends another **HTTP request** to the same server.

The browser finds the cookie for this server in its cookie file,

and attaches it to the request as a header line: **Cookie: id=1678**.

The server uses the ID as a key to look up the user's state in its database

Server can then generate a **tailored response** — e.g. a personalised page, a pre-filled shopping cart — for the client's request

HTTP

# One problem with the Web

Picture a university: thousands of users fetch the same popular pages and images, and every request travels across the Internet to the **origin server** that hosts the object.

Every one of those requests lands on the same origin server, through the same congested links — enough of them can **overwhelm** it.

**How can we avoid this?**

# Web caching

A **web cache** (also called a **proxy server**) is a network entity that satisfies HTTP requests on behalf of an origin server — storing copies of recently requested objects locally.




### Acts as a server

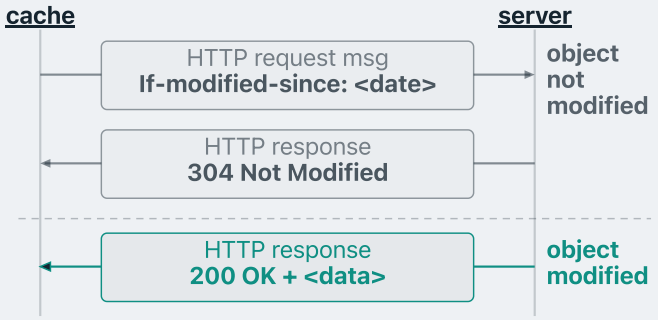
Receives HTTP requests from clients and, if it holds a copy of the object, returns it in the response — just like an origin server would

### Acts as a client

When the object is not cached, it issues its own HTTP request to the origin server, stores the copy, then answers the client

Requests answered by the cache get a **quicker response**, and caching **reduces how often** requests have to go to the origin server.

# Conditional GET




A **conditional GET** is an ordinary GET with an `If-modified-since` header line. A proxy uses it to check whether its cached copy is still up to date, e.g.:

```
GET /index.html HTTP/1.1
Host: www.example.com
If-modified-since: Mon, 10 Aug 2026 09:00:00 GMT
```

If the object is **unchanged**, server replies:

```
304 Not Modified
```

Object **changed**: server replies `200 OK` with the updated object in the body

MODULE 6

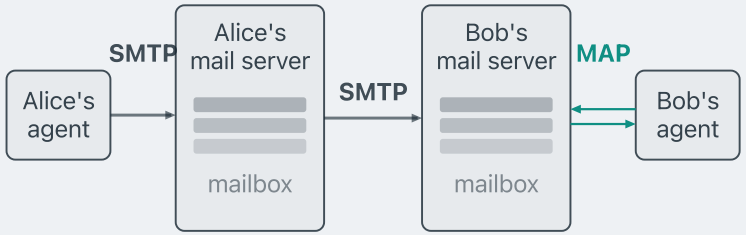
# Email

Electronic Mail

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EMAIL

# An overview of the email system




**User agent (UA)** — the app used to read, compose and send email (e.g. Outlook, Gmail, Apple Mail)

**Mail server** — a dedicated host that handles email on behalf of its users. It stores a **mailbox** per recipient for incoming mails and a **send queue** of outgoing mails

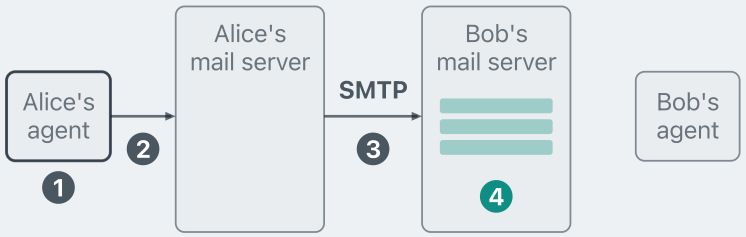
**SMTP** — Simple Mail Transfer Protocol; the sender's UA uses it to send a message to its own mail server

The sender's mail server then uses **SMTP** to transfer the message to the recipient's mail server across the internet

**Mail access protocol (MAP)** — used by UAs to retrieve messages from their mail server

# SMTP in action

SMTP has a **client side** and a **server side**, and always runs over **TCP**.




① Alice uses her UA to **compose** the message

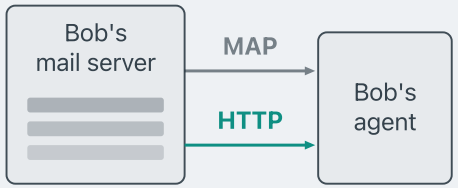
② Alice's UA (**SMTP client**) **sends** the message to her mail server (**SMTP server**)

③ The **SMTP client** on Alice's server opens a TCP connection to Bob's mail server and **transfers** Alice's message over it

④ The **SMTP server** on Bob's server receives the message and places it in Bob's **mailbox**

EMAIL

# Retrieving email from the server



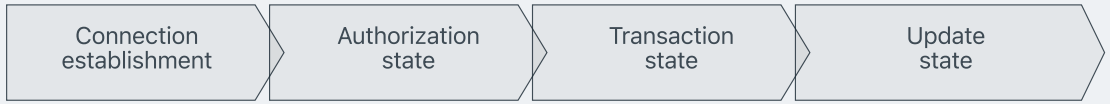

The recipient's UA retrieves email from its mail server either with a **mail access protocol (MAP)**, e.g.

- **POP3** — Post Office Protocol version 3 [RFC 1939]
- **IMAP** — Internet Message Access Protocol [RFC 9051]

or through **web-based email**, where the UA is a web browser that uses **HTTP** to reach the mail server

POP3

# How POP3 works




| Connection establishment | Authorization state | Transaction state | Update state |
|---|---|---|---|
| UA establishes a **TCP connection** to the mail server on **port 110** | UA sends **username** and **password** (cleartext) to authenticate | UA **retrieves messages**; can mark for deletion, remove marks, obtain statistics | Server **deletes** marked messages and **closes** the connection |

> Messages wait on the server until the UA downloads them

**Notable limitations:**

- Keeps no state on the server (e.g. read or unread), so what you do on one device is not seen on the others
- Does not support server-side folder management — users cannot organise emails into folders on the server

IMAP

# Internet Message Access Protocol

IMAP addresses POP3's limitations:

> IMAP **keeps per-message state on the server**, e.g. read or unread, flagged, so every device sees the same status

> It supports **folders on the server**: create folders and move messages into them, visible from any device

WEB-BASED EMAIL

# Accessing email via a browser

**Web-based email is accessed through a web browser**

The mailbox stays on the server; reading an email is like browsing the Web — the browser fetches each message as a web page

**Which application-layer protocol does the browser use to talk to the mail server?**

**HTTP**, the same protocol behind every other web page

**Web-based email services**

Examples: **Gmail**, **Outlook** (web)

EMAIL

# Comparing POP3, IMAP and web-based email

| | POP3 | IMAP | Web-based email |
|---|---|---|---|
| **Accessed with** | A mail app | A mail app | A web browser, over **HTTP** |
| **Messages** | Downloaded to the device; deleted from the server, or optionally kept | Kept on the server | Kept on the server |
| **Folders** | None on the server; a single mailbox | Created and kept on the server | Created and kept on the server |
| **Multiple devices** | Not synced: read status and folders exist only on each device | Read status and folders stay in sync across devices | Any device with a browser sees the same mailbox |

MESSAGE FORMAT

# Structure of an email message

An email message, as carried by **SMTP**, consists of **header fields**, followed (optionally) by a **body** [RFC 5322]

**Header fields** describe the message, e.g. `From:` and `Subject:`

The **body** carries the message content itself

SMTP requires the whole message to be **ASCII text**, which encodes English letters, digits and punctuation

**Problem:** ASCII alone cannot carry images, attachments, or non-ASCII characters (e.g. Chinese, Arabic)

**Header fields**

```
Date: Mon, 10 Aug 2026 09:00 GMT
From: alice@crepes.fr
To: bob@hamburger.edu
Subject: Picture of yummy crepe.
Cc: charlie@example.com
```

**Body**

```
Hello Bob, here is the picture
you asked for. Enjoy!
```

MIME

# Multipurpose Internet Mail Extensions

**MIME** extends the message format with **additional header fields**

These fields declare the content's **type** and how it is **encoded**

so the receiver knows how to **decode the body**

e.g. **base64** encoding converts binary data into printable ASCII text, which can be used for image attachments

```
From: alice@crepes.fr
To: bob@hamburger.edu
Subject: Picture of yummy crepe.
MIME-Version: 1.0
Content-Transfer-Encoding: base64
Content-Type: image/jpeg
---
(base64 encoded data…
…base64 encoded data)
```

→



decoded image

MODULE 6

# DNS

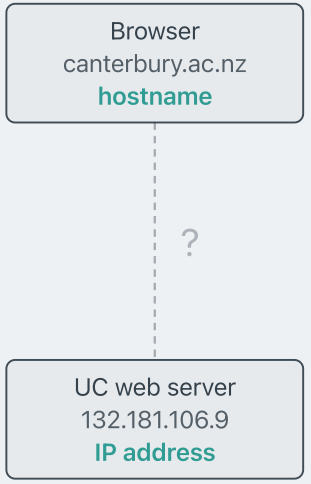
Domain Name System

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DNS

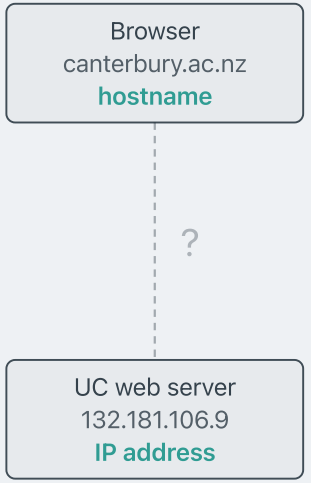
# Hostnames and IP addresses

- As we have seen, network routing entirely relies on **IP addresses** — numerical identifiers, e.g. `132.181.106.9`
- However, when you browse the web, you typically type a **human-friendly hostname** such as `canterbury.ac.nz`
- How does the browser know how to translate between a hostname and an IP address?



DNS

# Domain Name System — overview



DNS

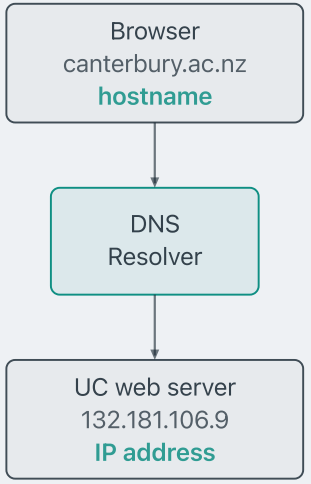
# Domain Name System — overview

**DNS** is that translation mechanism — it maps human-readable **hostnames** to numerical **IP addresses**

DNS stores this mapping in a **distributed database**, spread across a hierarchy of DNS servers [RFC 1034, 1035]

DNS is an **application-layer protocol** running mainly over **UDP on port 53** (TCP port 53 for large responses and zone transfers)

Why UDP? A query and its reply each fit in one small datagram, so there is no connection setup delay, providing a quick response.



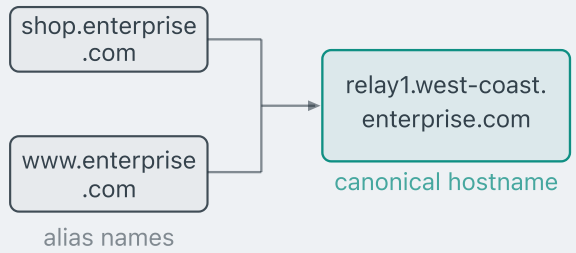

hostname → IP address

DNS SERVICES

# Additional services provided by DNS

**Host aliasing** — a host may have a cumbersome **canonical hostname**, e.g. `relay1.west-coast.enterprise.com`, which is hard to remember.

With host aliasing, it can also be reached via simpler **alias names**, e.g. `www.enterprise.com` and `shop.enterprise.com`; DNS maps each alias to the canonical hostname.



DNS SERVICES

# Additional services provided by DNS

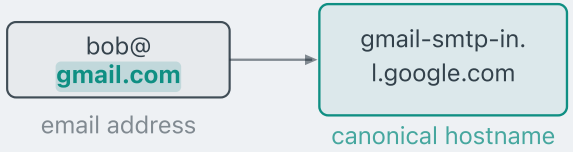
**Host aliasing** — a host may have a cumbersome **canonical hostname**, e.g. `relay1.west-coast.enterprise.com`, which is hard to remember.

With host aliasing, it can also be reached via simpler **alias names**, e.g. `www.enterprise.com` and `shop.enterprise.com`; DNS maps each alias to the canonical hostname.

**Mail server aliasing** — the same idea for email.

e.g. the mail server that handles Gmail has the cumbersome canonical hostname `gmail-smtp-in.l.google.com`.

Yet Bob's email address can simply be `bob@gmail.com`: DNS uses the part after the @, `gmail.com`, to find that mail server.



# Additional services by DNS: load distribution

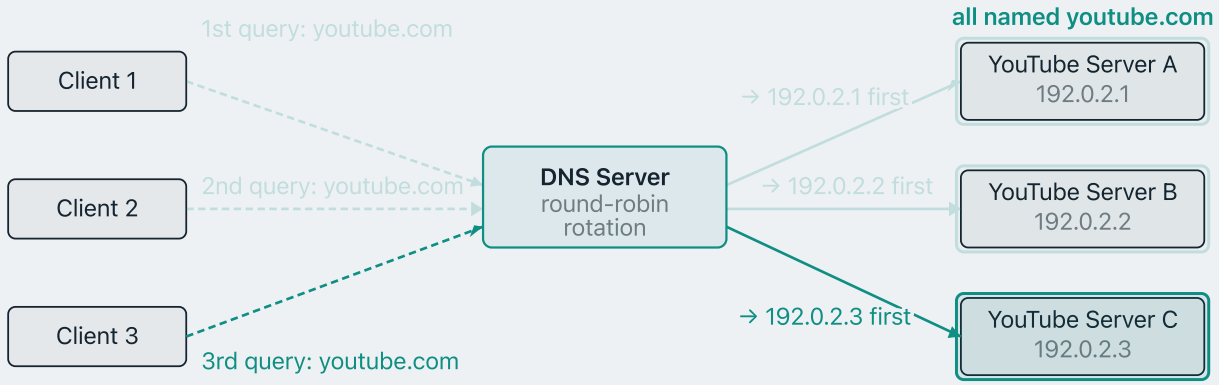
Many popular web services — e.g. YouTube, Google, Netflix — are hit by **millions of users** simultaneously. A **single server** cannot handle that volume.

The service is therefore distributed across **many server instances**, each with a **distinct IP address**, all sharing the **same hostname** (e.g. `youtube.com`).

### How are users assigned to these servers?

**DNS load balancing** — DNS stores all the IPs and **rotates their order** on each response, so consecutive clients get a different IP listed first.

e.g. three clients query `youtube.com` in turn:



DNS

# DNS is distributed, not centralised

Recall that DNS stores the hostname-to-IP mapping in a **distributed database**, spread across many DNS servers.

**Why not a centralised design — a single server hosting the whole database, so that every DNS query goes to it?**

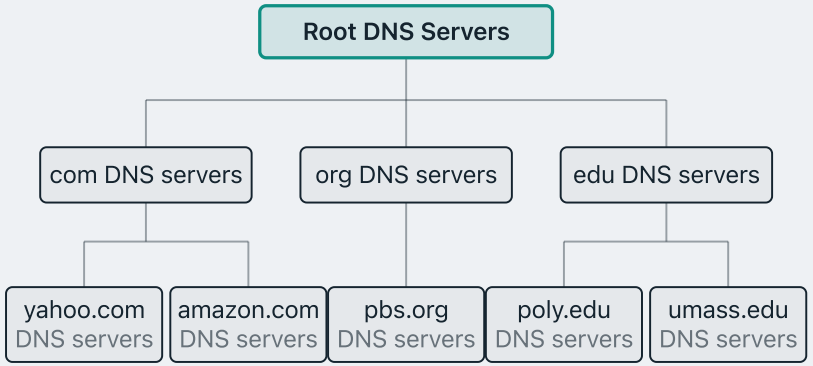
**Single point of failure**
If the one server crashes, DNS lookup fails for the entire Internet

**Traffic volume**
Billions of DNS queries per day — no single machine can handle that load

**Distant centralised database**
All queries must travel to one location — users far away face high latency on every lookup

**Maintenance**
Billions of hostname records change constantly — impossible to keep one database current

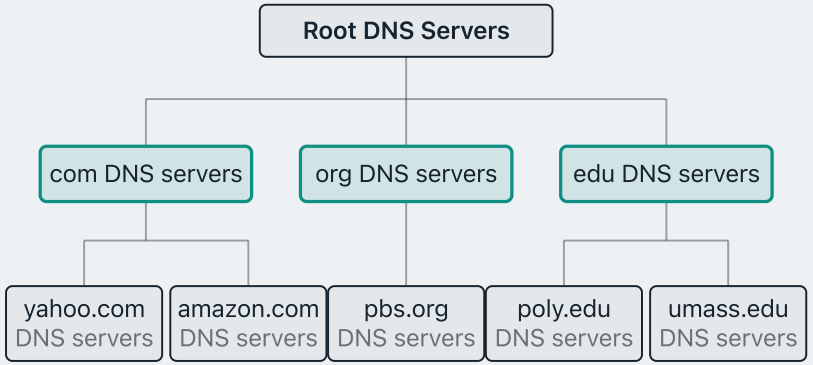
# A hierarchical distributed database




**Root DNS Servers**

- The top of the hierarchy: looking up a hostname from scratch **starts here**
- There are 13 root server identities (a–m.root-servers.net)
- Each is replicated: its IP address is shared by many machines worldwide, and a query goes to the nearest one

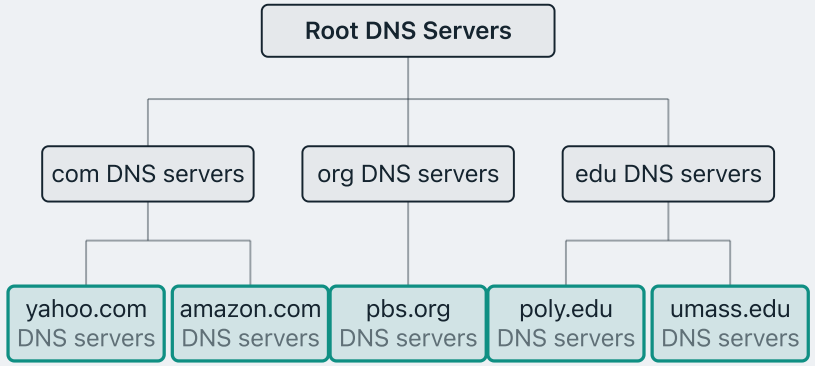
# A hierarchical distributed database




**Top-Level Domain (TLD) DNS Servers**

- Handle generic domains: **.com**, **.org**, **.edu**
- ...and country-code domains: **.nz**, **.uk**
- Each is run by a dedicated registry, e.g. **Verisign** (.com), **InternetNZ** (.nz)

# A hierarchical distributed database




**Authoritative DNS Servers**

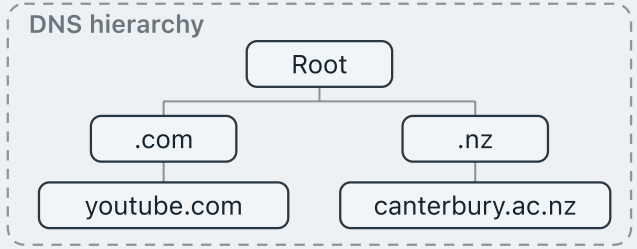
- Each organisation's own DNS server
- It stores the organisation's own records — the original, **definitive** copy that every answer for its hostnames comes from
- e.g. Canterbury's authoritative server holds the IP for `www.canterbury.ac.nz`

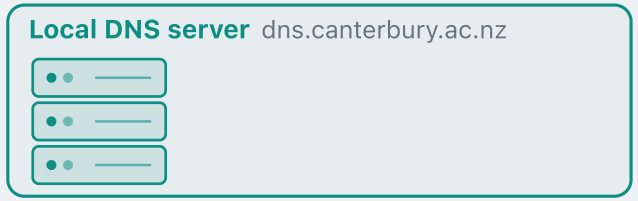
DNS ARCHITECTURE

# Local DNS server and DNS caching

Every ISP or institution operates a **local DNS server** (also called a *default name server*).

It is **not** part of the root / TLD / authoritative hierarchy — it sits **separately** from that.

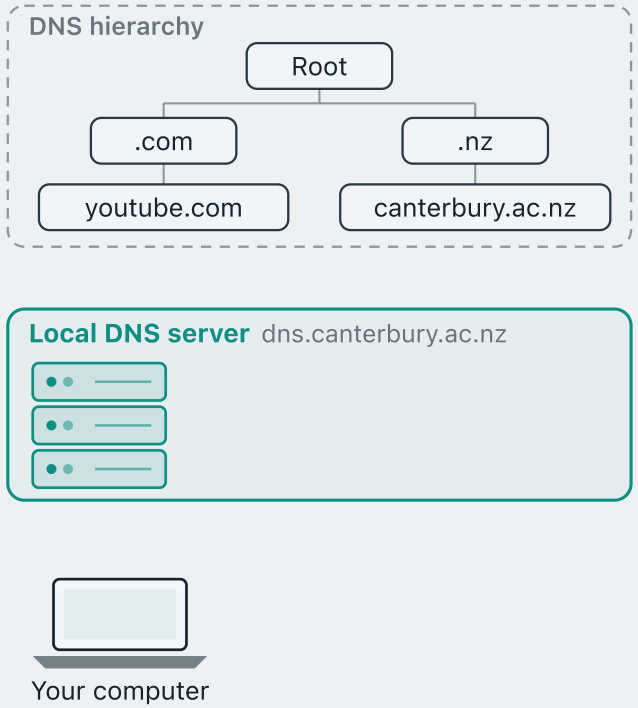


DNS ARCHITECTURE

# Local DNS server and DNS caching

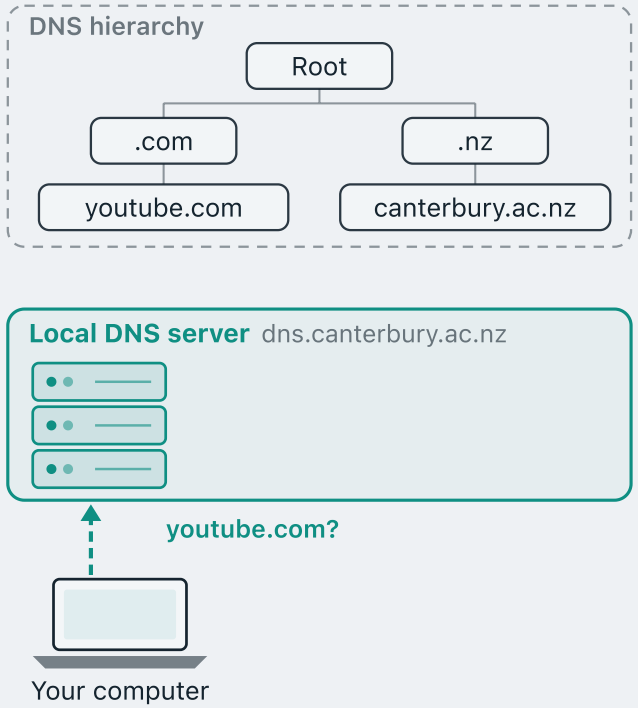
When you join the network, your device is automatically configured to use one — e.g. the University of Canterbury runs its own local DNS server for all campus users.



DNS ARCHITECTURE

# Local DNS server and DNS caching

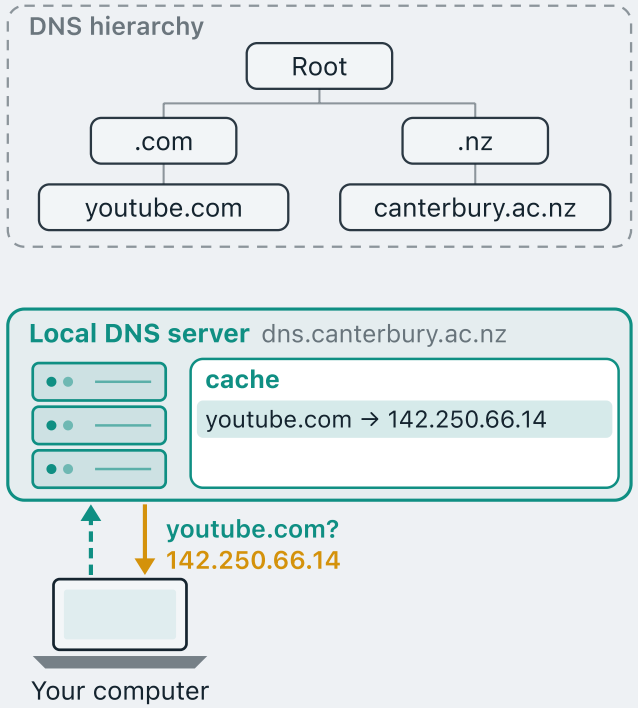
When your machine makes a DNS lookup, the query goes **first to the local DNS server**.



DNS ARCHITECTURE

# Local DNS server and DNS caching

**If the answer is cached:** the local DNS server replies **immediately** — no hierarchy lookup needed.



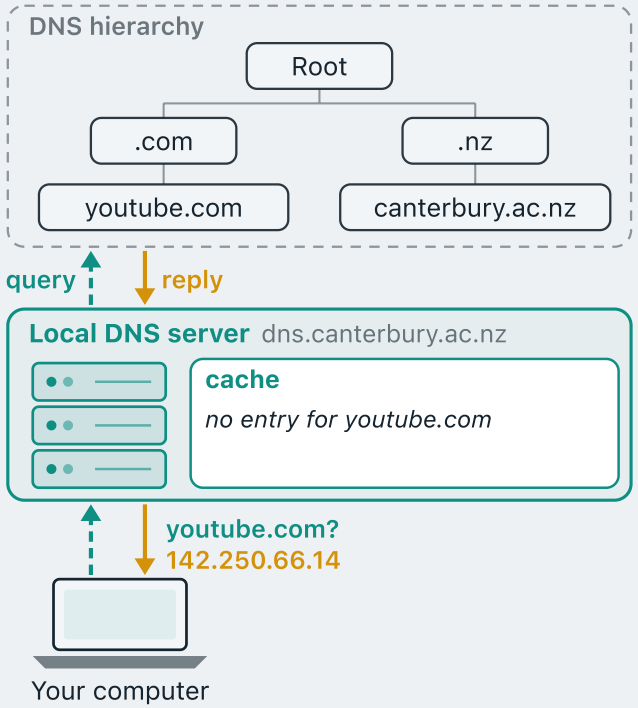
DNS ARCHITECTURE

# Local DNS server and DNS caching

**If not cached:** the local server **forwards the query into the hierarchy**.

The hierarchy replies with the IP address.

The local server **relays the reply** back to you,



DNS ARCHITECTURE

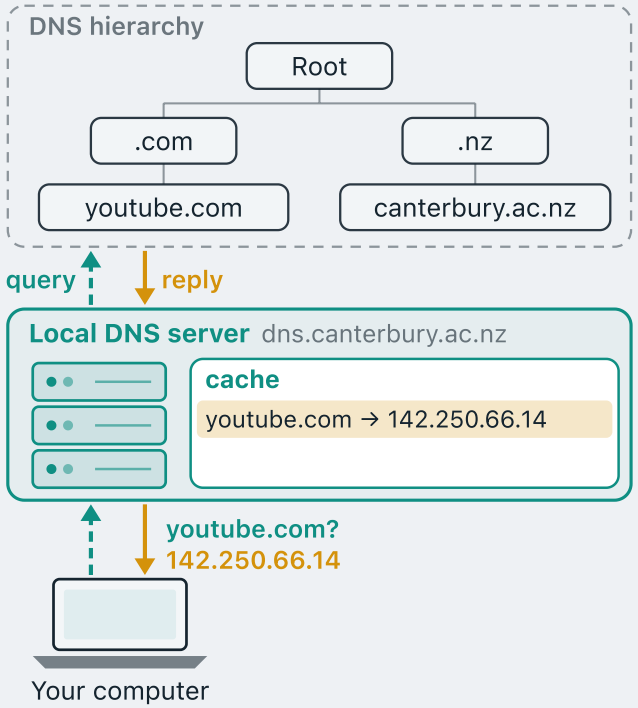
# Local DNS server and DNS caching

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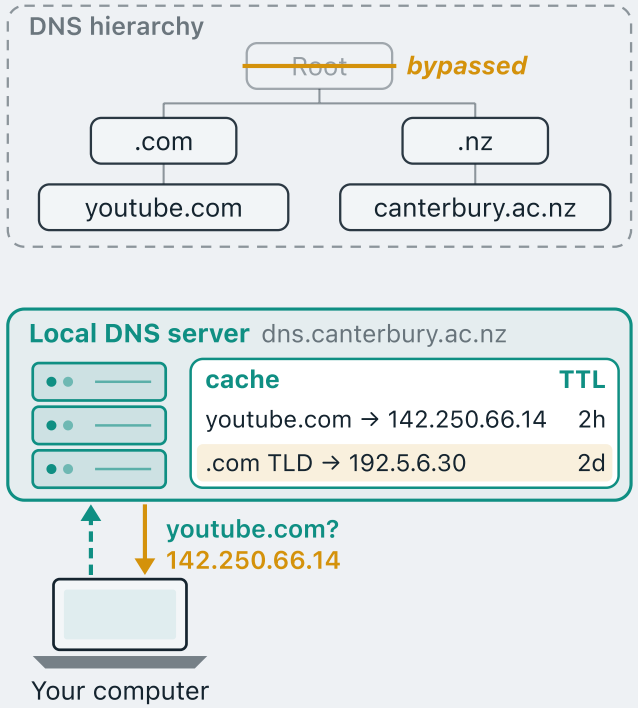
and **caches the hostname-to-IP mapping** for future queries.



DNS ARCHITECTURE

# Local DNS server and DNS caching

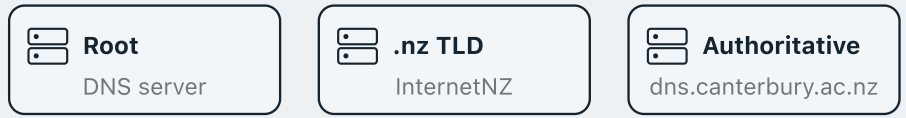
Cache entries have a **TTL (time-to-live)** and expire after some time (typically hours). TLD server addresses are routinely cached in local DNS servers — so **root servers are bypassed for most everyday queries**.

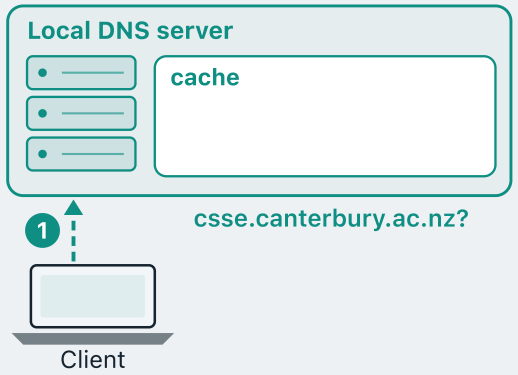


DNS ARCHITECTURE

# Resolving a hostname: the full journey

**Step 1.** A client wants to resolve **csse.canterbury.ac.nz**. It sends a DNS query to its **local DNS server**.

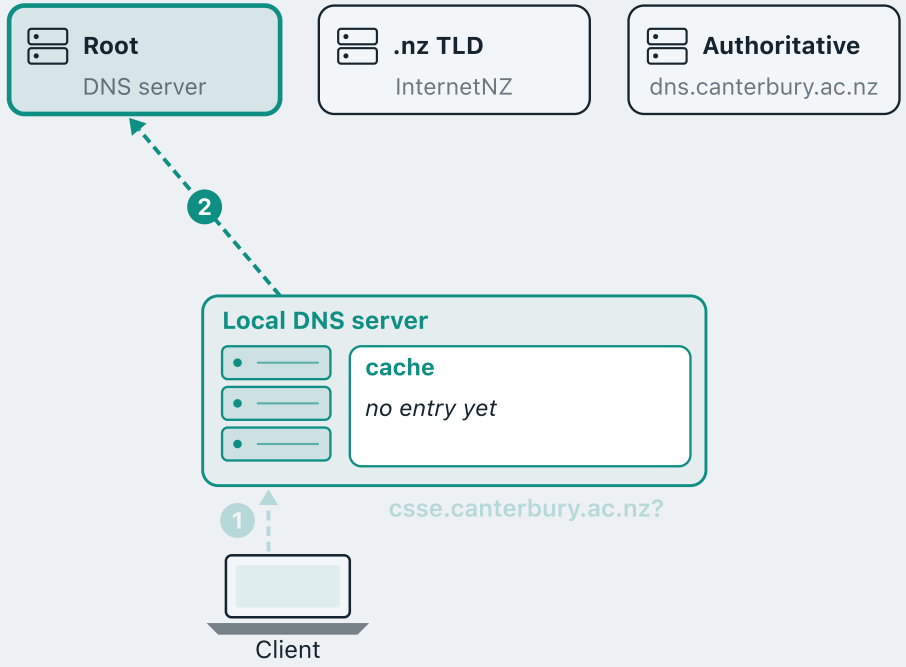


DNS ARCHITECTURE

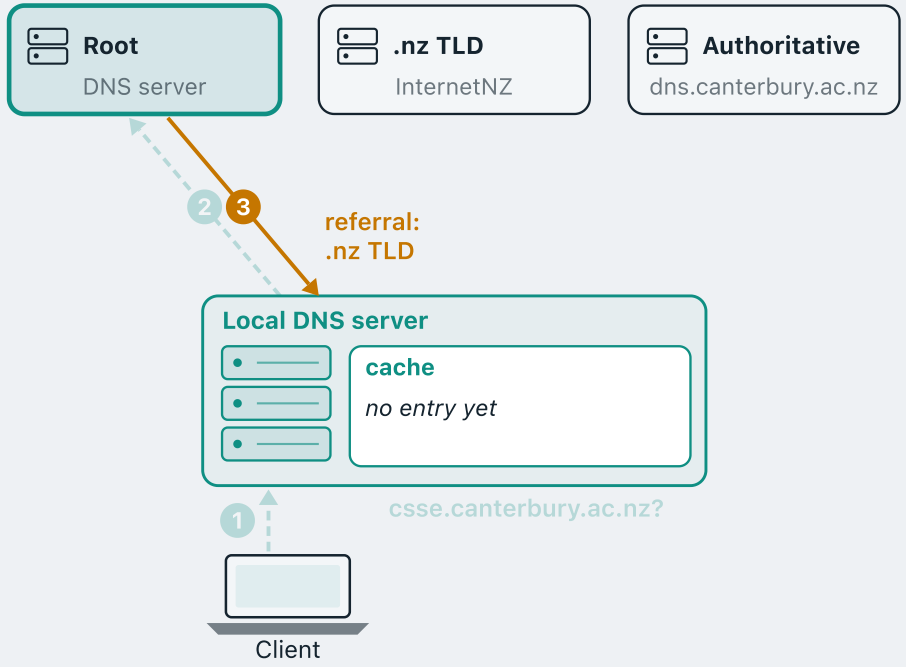
# Resolving a hostname: the full journey

**Step 2.** The local DNS doesn't have the answer cached. It contacts a **root DNS server**.



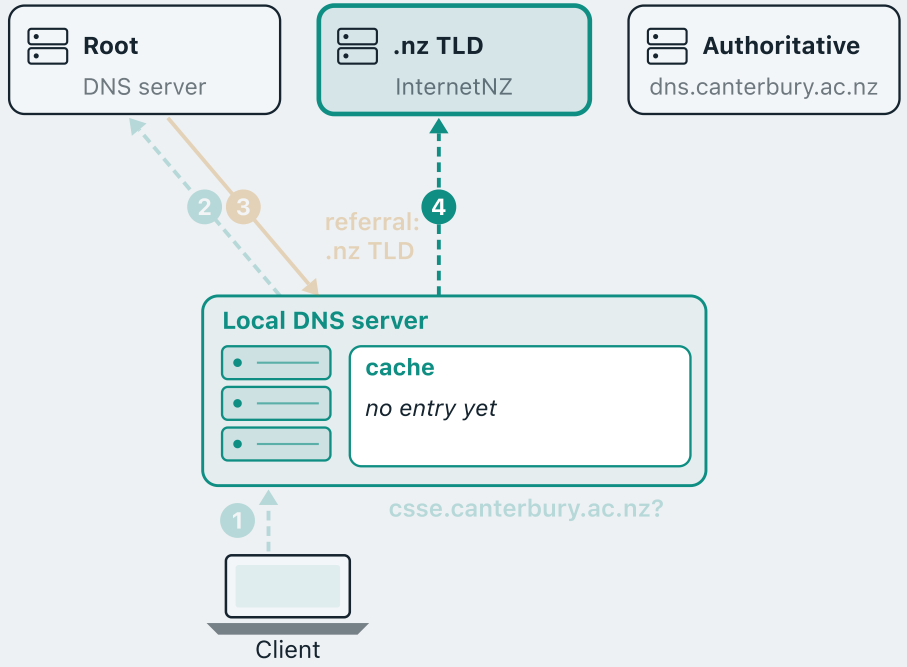
# Resolving a hostname: the full journey

**Step 3.** The root server returns a **referral**: "I don't know — try a **.nz TLD** server" and gives its address.



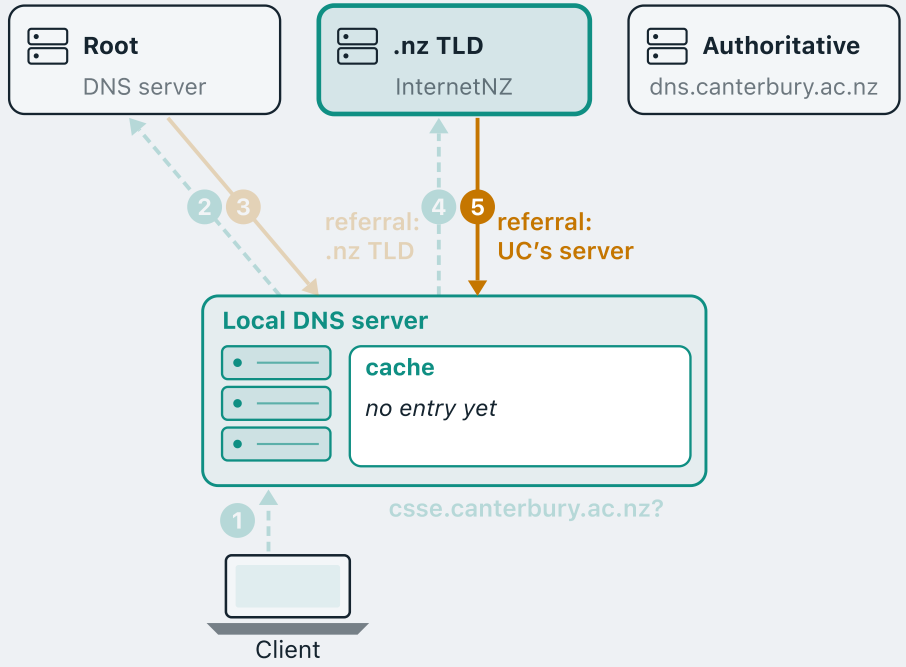
# Resolving a hostname: the full journey

**Step 4.** The local DNS follows the referral and contacts the **.nz TLD server**.



# Resolving a hostname: the full journey

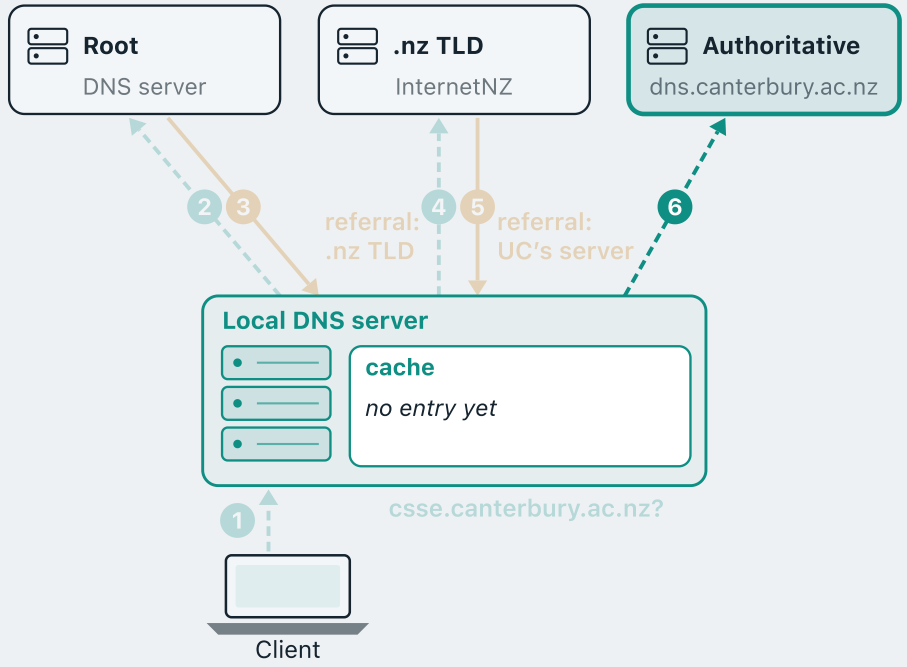
**Step 5.** The .nz TLD returns another **referral**: "Try **dns.canterbury.ac.nz** — the authoritative server for canterbury.ac.nz."



DNS ARCHITECTURE

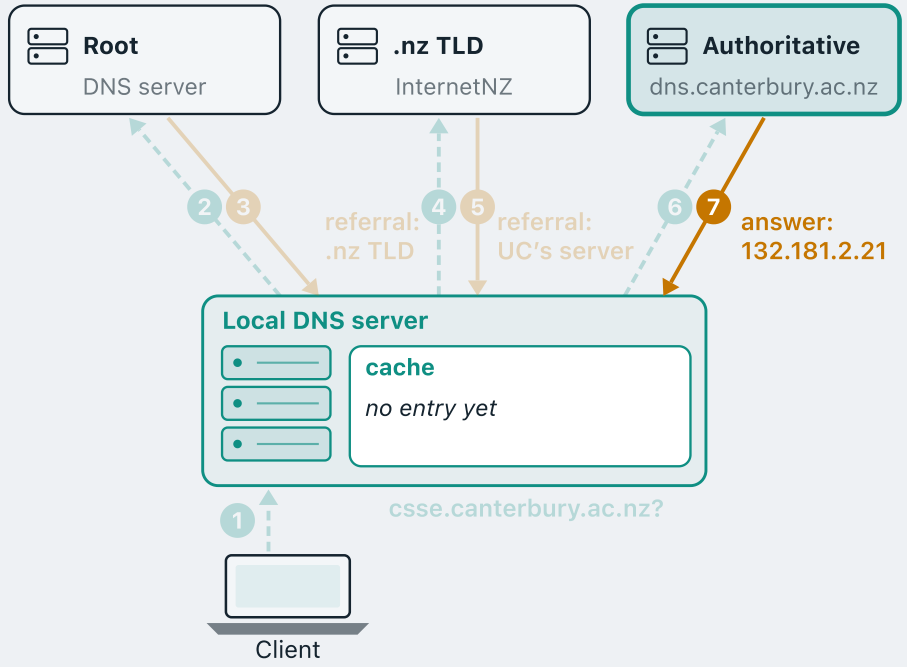
# Resolving a hostname: the full journey

**Step 6.** The local DNS contacts **dns.canterbury.ac.nz** — the authoritative name server for the canterbury.ac.nz zone.



# Resolving a hostname: the full journey

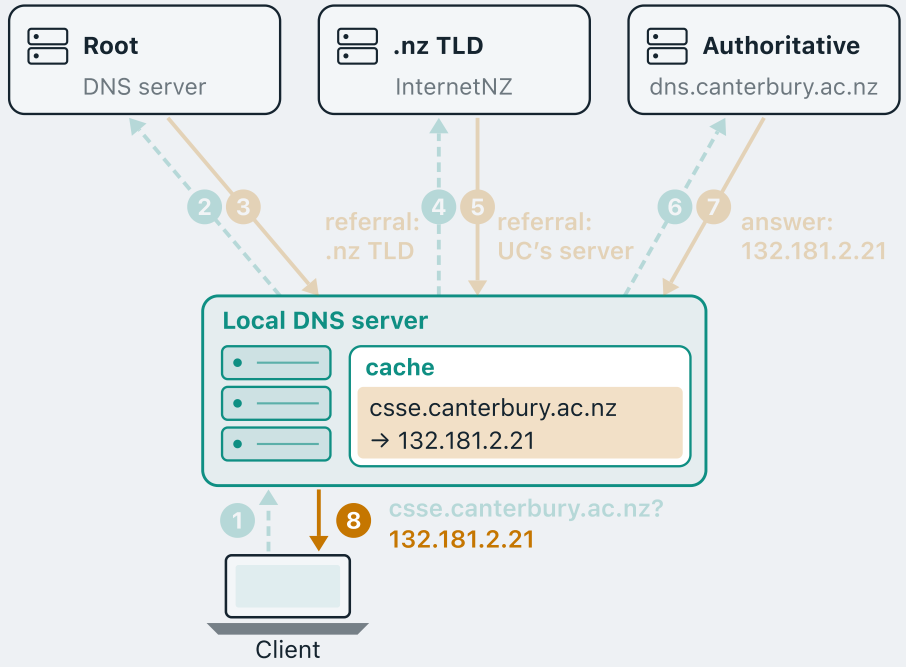
**Step 7.** The authoritative server has the answer! It replies: "csse.canterbury.ac.nz = **132.181.2.21**"



DNS ARCHITECTURE

# Resolving a hostname: the full journey

**Step 8.** The local DNS returns **132.181.2.21** to the client — and **caches** the mapping for future queries.

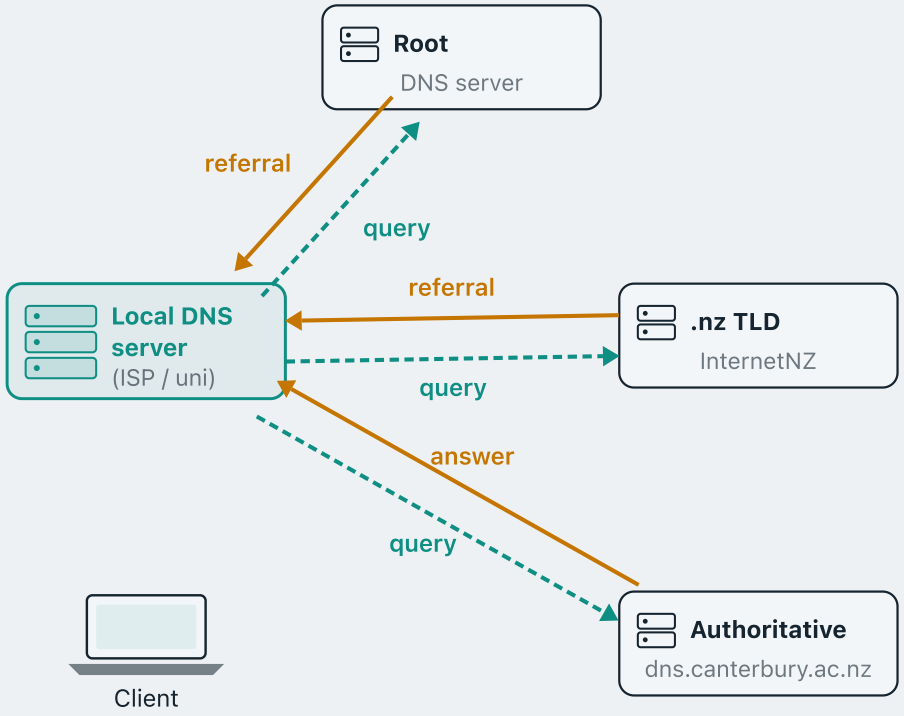


DNS ARCHITECTURE

# Recursive vs iterative queries

### Iterative query

- The contacted server does **not** query other servers for you: it replies only with what it knows itself.
- If it doesn't know the answer, it returns a **referral**: "I don't know — try that server."
- The authoritative server *does* know, so its reply to the same kind of query is the **answer**.

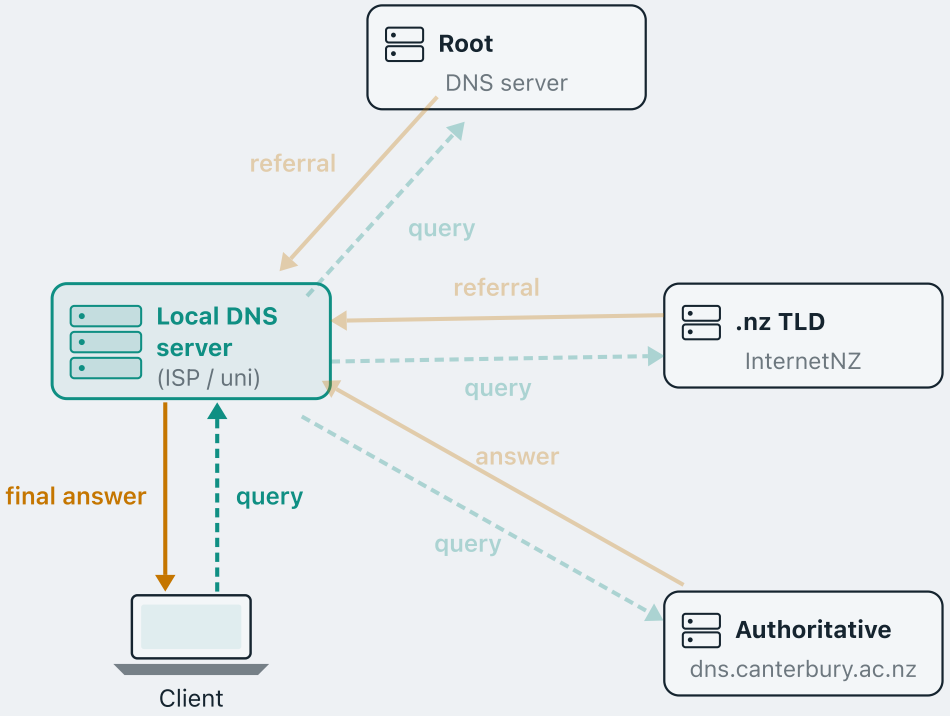


DNS ARCHITECTURE

# Recursive vs iterative queries

**Recursive query**

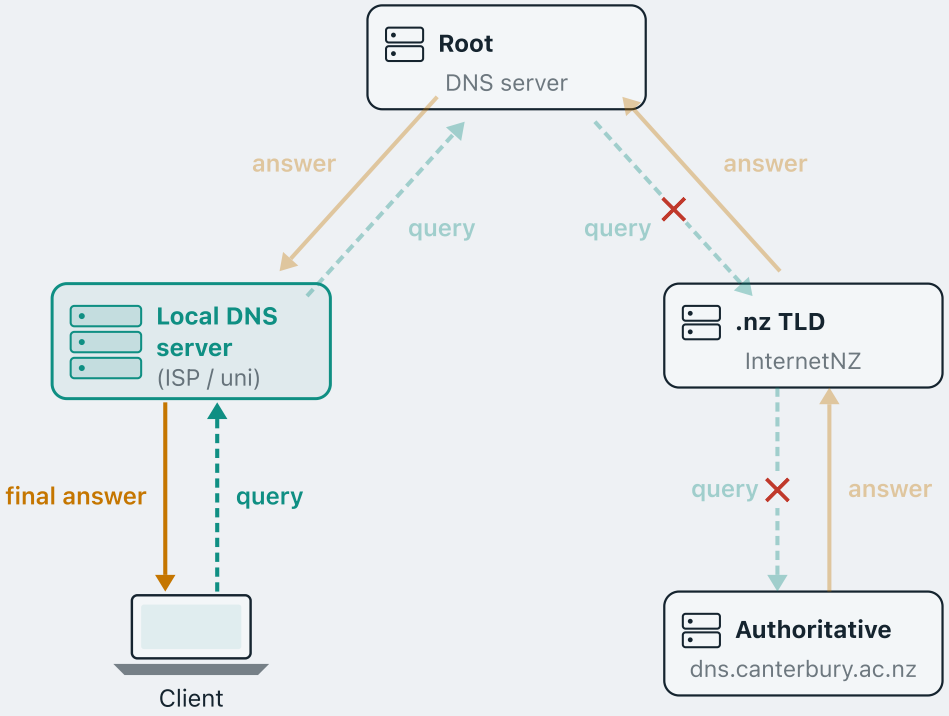
- The contacted server takes **full responsibility**:
- it resolves the name itself, querying the next servers on your behalf,
- and returns the **final answer**. e.g. client → local DNS server.



# Recursive vs iterative queries

**Can we go fully recursive?**

- In principle, yes: root asks the TLD server, which asks the authoritative server,
- and the answer flows back along the chain.
- In practice, root and TLD servers **refuse recursion**: too much load.



DNS ARCHITECTURE

# DNS Resource Records

DNS's distributed database stores its mappings as **Resource Records (RRs)**: each DNS server keeps a table of records, all in the same four-field format.

| **name** | **value** | **type** | **TTL** |
|---|---|---|---|
| e.g. a hostname | e.g. an IP address | A, NS, CNAME, MX | time-to-live (s) |

**Type A**

**name** = hostname · **value** = IPv4 address

| NAME | VALUE | TYPE | TTL (s) |
|---|---|---|---|
| web.canterbury.ac.nz | 132.181.106.9 | A | 3600 |
| itdns.canterbury.ac.nz | 132.181.2.225 | A | 3600 |
| mail.canterbury.ac.nz | 132.181.3.225 | A | 3600 |
| canterbury.ac.nz | itdns.canterbury.ac.nz | NS | 86400 |
| www.canterbury.ac.nz | web.canterbury.ac.nz | CNAME | 3600 |
| canterbury.ac.nz | mail.canterbury.ac.nz | MX | 3600 |

DNS ARCHITECTURE

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| **name** | **value** | **type** | **TTL** |
|---|---|---|---|
| e.g. a hostname | e.g. an IP address | A, NS, CNAME, MX | time-to-live (s) |

### Type NS

**name** = domain · **value** = hostname of the authoritative DNS server for that domain

*"I don't know — but here's the name of the server that does."*

| NAME | VALUE | TYPE | TTL (s) |
|---|---|---|---|
| web.canterbury.ac.nz | 132.181.106.9 | A | 3600 |
| itdns.canterbury.ac.nz | 132.181.2.225 | A | 3600 |
| mail.canterbury.ac.nz | 132.181.3.225 | A | 3600 |
| canterbury.ac.nz | itdns.canterbury.ac.nz | NS | 86400 |
| www.canterbury.ac.nz | web.canterbury.ac.nz | CNAME | 3600 |
| canterbury.ac.nz | mail.canterbury.ac.nz | MX | 3600 |

DNS ARCHITECTURE

# DNS Resource Records

DNS's distributed database stores its mappings as **Resource Records (RRs)**: each DNS server keeps a table of records, all in the same four-field format.

- **name** — e.g. a hostname
- **value** — e.g. an IP address
- **type** — A, NS, CNAME, MX
- **TTL** — time-to-live (s)

### Type CNAME

**name** = alias hostname · **value** = canonical (real) hostname it maps to

| NAME | VALUE | TYPE | TTL (s) |
|---|---|---|---|
| web.canterbury.ac.nz | 132.181.106.9 | A | 3600 |
| itdns.canterbury.ac.nz | 132.181.2.225 | A | 3600 |
| mail.canterbury.ac.nz | 132.181.3.225 | A | 3600 |
| canterbury.ac.nz | itdns.canterbury.ac.nz | NS | 86400 |
| www.canterbury.ac.nz | web.canterbury.ac.nz | CNAME | 3600 |
| canterbury.ac.nz | mail.canterbury.ac.nz | MX | 3600 |

DNS ARCHITECTURE

# DNS Resource Records

DNS's distributed database stores its mappings as **Resource Records (RRs)**: each DNS server keeps a table of records, all in the same four-field format.

| **name** | **value** | **type** | **TTL** |
|---|---|---|---|
| e.g. a hostname | e.g. an IP address | A, NS, CNAME, MX | time-to-live (s) |

### Type MX

**name** = domain · **value** = hostname of the mail server for that domain

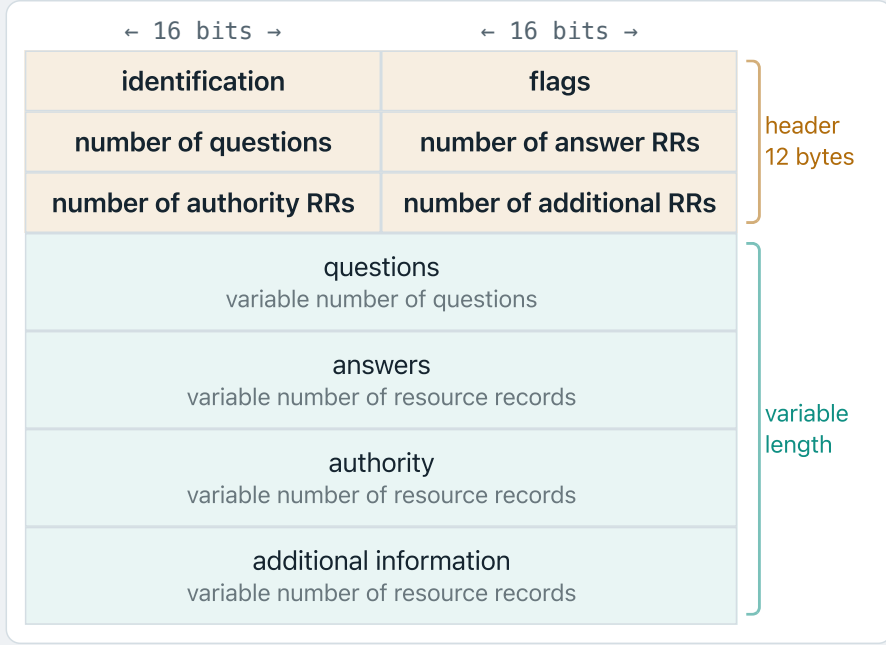
Unlike CNAME, it only redirects mail: a sending mail server (**SMTP**) looks it up to find where to deliver a domain's email.

| NAME | VALUE | TYPE | TTL (s) |
|---|---|---|---|
| web.canterbury.ac.nz | 132.181.106.9 | A | 3600 |
| itdns.canterbury.ac.nz | 132.181.2.225 | A | 3600 |
| mail.canterbury.ac.nz | 132.181.3.225 | A | 3600 |
| canterbury.ac.nz | itdns.canterbury.ac.nz | NS | 86400 |
| www.canterbury.ac.nz | web.canterbury.ac.nz | CNAME | 3600 |
| canterbury.ac.nz | mail.canterbury.ac.nz | MX | 3600 |

DNS ARCHITECTURE

# DNS protocol message format

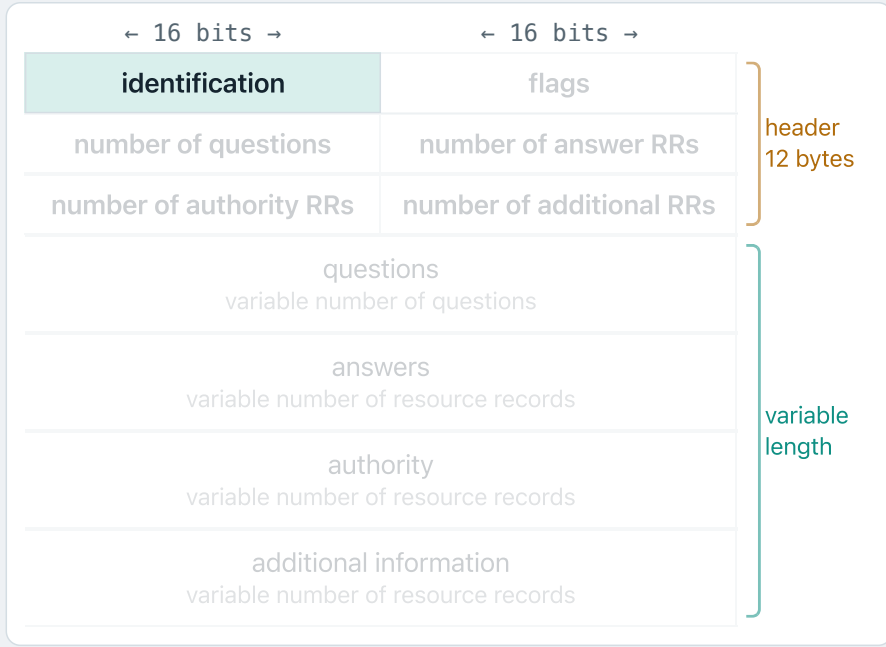
DNS uses the **same message format** for both **queries** and **replies** — a fixed 12-byte header followed by four variable-length sections.



# DNS protocol message format

**Identification**

16-bit number assigned by the client. The reply copies this same value, so the client can match each reply to its original query.

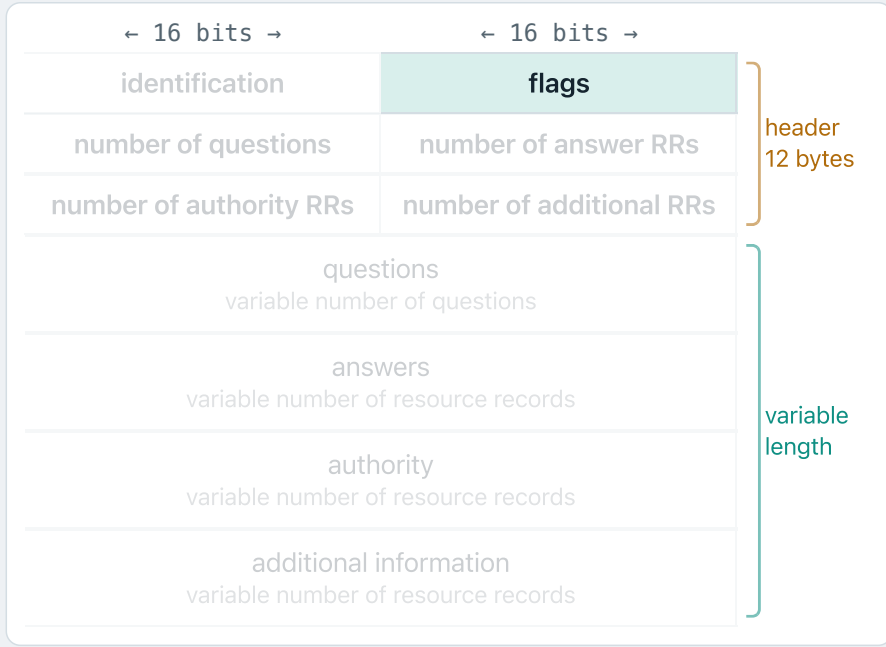


DNS ARCHITECTURE

# DNS protocol message format

**Flags**

- Query (0) or reply (1)
- Recursion desired (RD) — client requests full recursive resolution
- Recursion available (RA) — server indicates it supports recursion

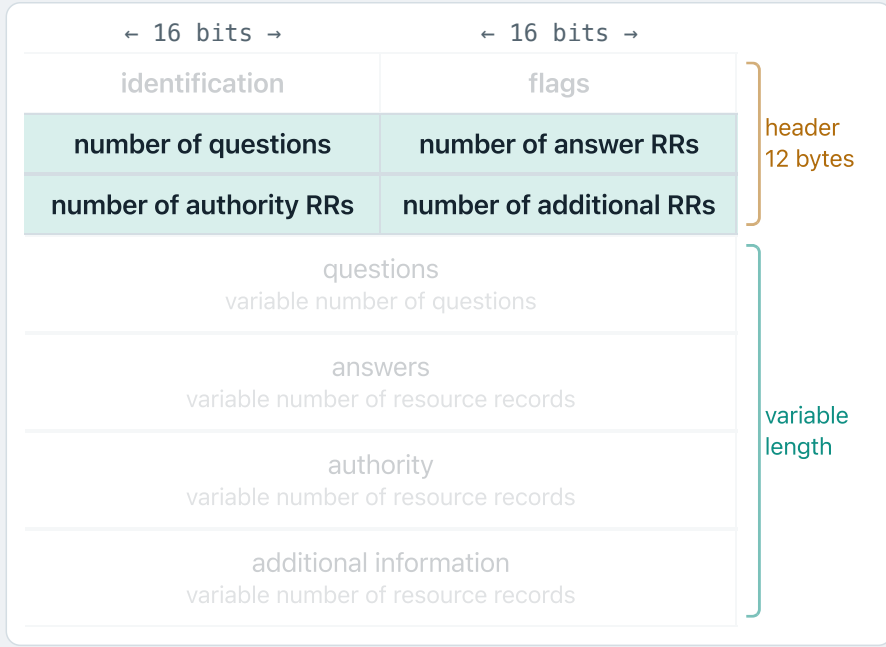


DNS ARCHITECTURE

# DNS protocol message format

### Count fields

Four 16-bit counters — one per section below — telling how many records appear in each variable-length section.

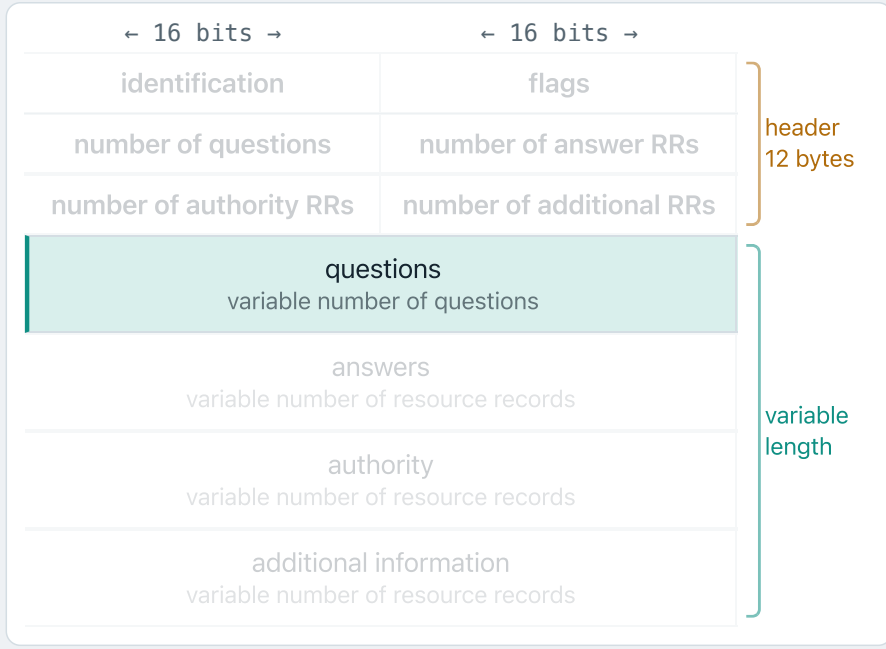


# DNS protocol message format

### Questions

The name and record type being queried. e.g. this Type A question asks for the IP address of the domain facebook.com.

```
(facebook.com, A)
```

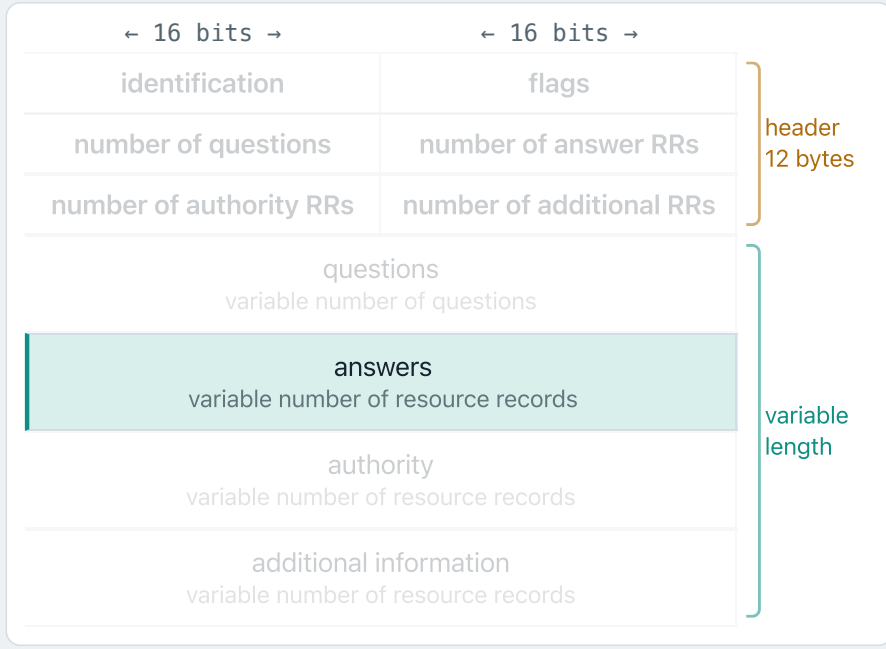


# DNS protocol message format

**Answers**

Resource Records (RRs) that directly answer the query.

```
(facebook.com, 157.240.22.35, A, TTL=300)
```

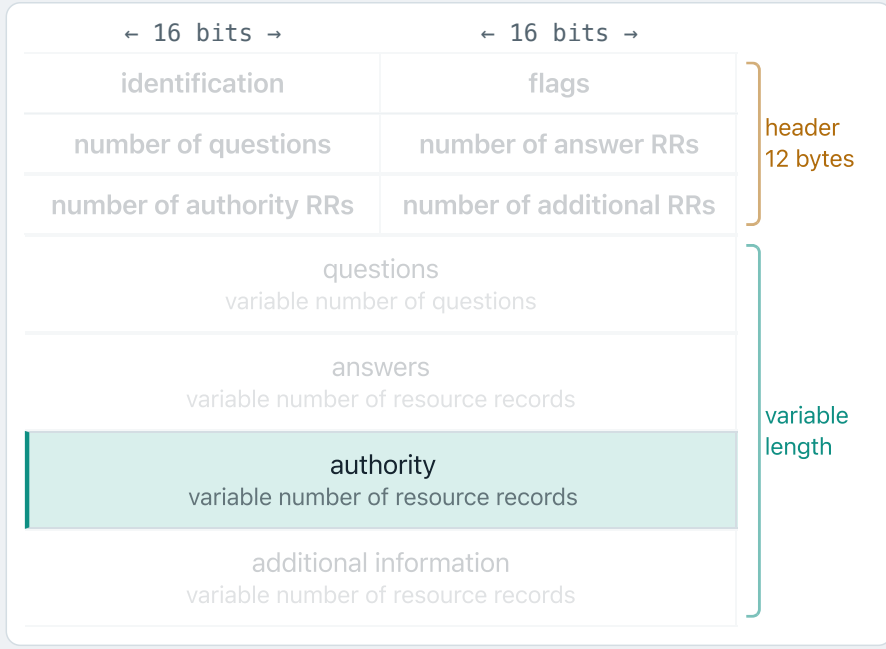


# DNS protocol message format

**Authority**

NS records pointing to authoritative servers — returned in referral responses.

```
(facebook.com, a.ns.facebook.com, NS)
```

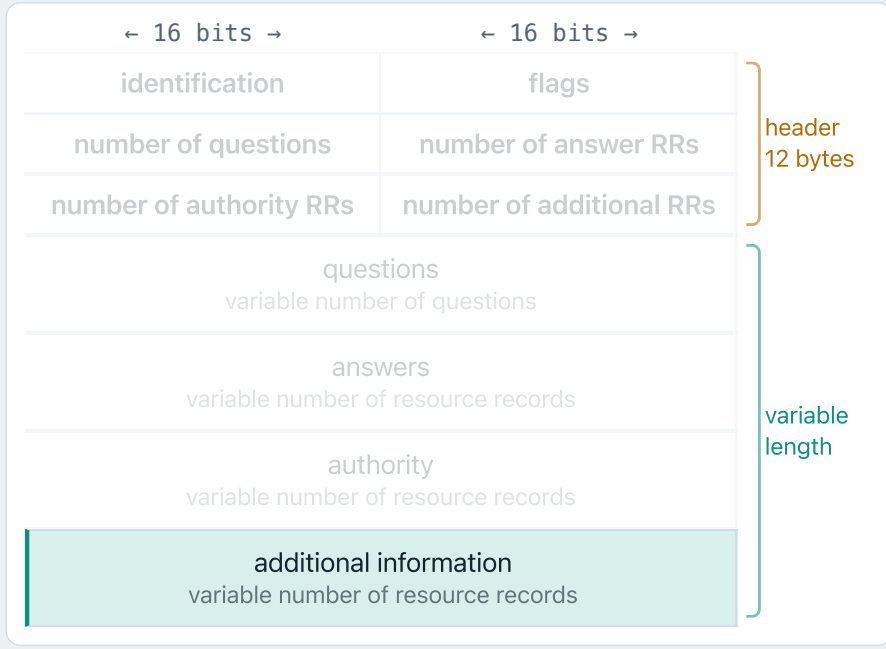


# DNS protocol message format

**Additional**

'Glue' Resource Records (RRs) accompanying the authority section — so you can reach the name server without an extra lookup.

```
(a.ns.facebook.com, 129.134.30.12, A)
```



# Query & response: an example

**QUERY** — client → local DNS · UDP 53

```
Transaction ID  0x6492
Flags           0x0100 (query, RD=1)
Questions  1    Answer RRs  0
Authority RRs  0    Additional RRs  0
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

### The question

The query asks one question: what is the IP address (type **A**) of `www.canterbury.ac.nz`?

**RESPONSE** — local DNS → client

```
Transaction ID  0x6492
Flags           0x8180 (reply, RA=1, no error)
Questions  1    Answer RRs  2
Authority RRs  1    Additional RRs  1
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**ANSWERS**

```
www.canterbury.ac.nz    CNAME
  web.canterbury.ac.nz
web.canterbury.ac.nz    A
  132.181.106.9
```

**AUTHORITY**

```
canterbury.ac.nz    NS
  itdns.canterbury.ac.nz
```

**ADDITIONAL**

```
itdns.canterbury.ac.nz    A
  132.181.2.225
```

# Query & response: an example

**QUERY** client → local DNS · UDP 53

```
Transaction ID  0x6492
Flags           0x0100 (query, RD=1)
Questions  1    Answer RRs  0
Authority RRs  0    Additional RRs  0
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**The answer**

The response answers with **two records**.

**RESPONSE** local DNS → client

```
Transaction ID  0x6492
Flags           0x8180 (reply, RA=1, no error)
Questions  1    Answer RRs  2
Authority RRs  1    Additional RRs  1
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**ANSWERS**

```
www.canterbury.ac.nz    CNAME
  web.canterbury.ac.nz
web.canterbury.ac.nz    A
  132.181.106.9
```

**AUTHORITY**

```
canterbury.ac.nz    NS
  itdns.canterbury.ac.nz
```

**ADDITIONAL**

```
itdns.canterbury.ac.nz    A
  132.181.2.225
```

# Query & response: an example

**QUERY** client → local DNS · UDP 53

```
Transaction ID  0x6492
Flags           0x0100 (query, RD=1)
Questions  1    Answer RRs  0
Authority RRs  0    Additional RRs  0
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**First: a CNAME**

`www.canterbury.ac.nz` is an **alias**; its canonical name is `web.canterbury.ac.nz`.

**RESPONSE** local DNS → client

```
Transaction ID  0x6492
Flags           0x8180 (reply, RA=1, no error)
Questions  1    Answer RRs  2
Authority RRs  1    Additional RRs  1
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**ANSWERS**

```
www.canterbury.ac.nz    CNAME
  web.canterbury.ac.nz
web.canterbury.ac.nz    A
  132.181.106.9
```

**AUTHORITY**

```
canterbury.ac.nz    NS
  itdns.canterbury.ac.nz
```

**ADDITIONAL**

```
itdns.canterbury.ac.nz    A
  132.181.2.225
```

# Query & response: an example

**QUERY** client → local DNS · UDP 53

```
Transaction ID  0x6492
Flags           0x0100 (query, RD=1)
Questions  1    Answer RRs  0
Authority RRs  0    Additional RRs  0
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**Then: an A record**

The canonical name's IP address, 132.181.106.9 — the answer we wanted.

**RESPONSE** local DNS → client

```
Transaction ID  0x6492
Flags           0x8180 (reply, RA=1, no error)
Questions  1    Answer RRs  2
Authority RRs  1    Additional RRs  1
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**ANSWERS**

```
www.canterbury.ac.nz    CNAME
  web.canterbury.ac.nz
web.canterbury.ac.nz    A
  132.181.106.9
```

**AUTHORITY**

```
canterbury.ac.nz    NS
  itdns.canterbury.ac.nz
```

**ADDITIONAL**

```
itdns.canterbury.ac.nz    A
  132.181.2.225
```

# Query & response: an example

**QUERY** client → local DNS · UDP 53

```
Transaction ID  0x6492
Flags           0x0100 (query, RD=1)
Questions  1    Answer RRs  0
Authority RRs  0    Additional RRs  0
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

### Authority

An **NS** record naming an authoritative server for `canterbury.ac.nz`: the final level, which holds the answer's record.

**RESPONSE** local DNS → client

```
Transaction ID  0x6492
Flags           0x8180 (reply, RA=1, no error)
Questions  1    Answer RRs  2
Authority RRs  1    Additional RRs  1
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**ANSWERS**

```
www.canterbury.ac.nz    CNAME
  web.canterbury.ac.nz
web.canterbury.ac.nz    A
  132.181.106.9
```

**AUTHORITY**

```
canterbury.ac.nz    NS
  itdns.canterbury.ac.nz
```

**ADDITIONAL**

```
itdns.canterbury.ac.nz    A
  132.181.2.225
```

# Query & response: an example

**QUERY** — client → local DNS · UDP 53

```
Transaction ID  0x6492
Flags           0x0100 (query, RD=1)
Questions  1    Answer RRs  0
Authority RRs  0    Additional RRs  0
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**Additional**

The **IP address** of that name server, so it can be reached directly.

**RESPONSE** — local DNS → client

```
Transaction ID  0x6492
Flags           0x8180 (reply, RA=1, no error)
Questions  1    Answer RRs  2
Authority RRs  1    Additional RRs  1
```

**QUESTIONS**

```
www.canterbury.ac.nz    A
```

**ANSWERS**

```
www.canterbury.ac.nz    CNAME
  web.canterbury.ac.nz
web.canterbury.ac.nz    A
  132.181.106.9
```

**AUTHORITY**

```
canterbury.ac.nz    NS
  itdns.canterbury.ac.nz
```

**ADDITIONAL**

```
itdns.canterbury.ac.nz    A
  132.181.2.225
```

DNS

# DNS — key takeaways

- **Main purpose** — DNS maps human-readable hostnames to IP addresses.
- **Hierarchy** — DNS is distributed: no single server holds all records.
  Responsibility is delegated across Root, TLD, and Authoritative servers.
- **Traversal pattern** — the client queries the local DNS resolver, which then follows referrals Root → TLD → Authoritative.
  Results are cached locally by TTL to avoid repeated lookups.

# References


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William Stallings, *Data and Computer Communications*, Pearson, 10th edition, 2013.

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Andrew S. Tanenbaum & David J. Wetherall, *Computer Networks*, Prentice Hall, 5th edition, 2010.

Chapter 7 The Application Layer (7.1–7.3)

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